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Chemical cross-linking: A feasible approach to prolong doxylamine/pyridoxine release from spray-dried chitosan microspheres

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ABSTRACT

Spray-dried chitosan microparticles have been widely exploited as vehicles for mucosal drug delivery. Despite their advantages as pharmaceutical formulations, one of the major challenges is achieving sustained drug release, which would diminish toxicity and dosage frequency. The aim of this study was to formulate mucoadhesive glutaraldehyde cross-linked chitosan microparticles loaded with doxylamine succinate and pyridoxine hydrochloride as potential nasal drug delivery systems with sustained release. Microparticle models were formulated *via* spray-drying technique, using glutaraldehyde in different concentrations (0.1–1.0 mg/mL) as a cross-linking agent for chitosan. The obtained particles were with spherical shape, smooth surface and median diameter of 4 μm. The drug entrapment efficiency was high (80.47%–94.25%), indicating a tendency to decrease at higher glutaraldehyde concentrations. FTIR data demonstrated that there were no chemical interactions between glutaraldehyde and the drugs. The *in vitro* studies showed that the cross-linking process substantially limited particles swelling. The cross-linked particles exhibited sustained drug release characteristics at pH 6.8 over a period of 5 h with an initial burst-effect in the first 30 min. Drug release followed Korsmeyer-Peppas kinetics. Although a decrease of the particles mucoadhesive properties was observed after modification, all cross-linked formulations demonstrated high *in vitro* adsorption of mucin. The proposed models of mucoadhesive microsphere with sustained drug release are a perspective ground for further development of a novel delivery system for nasal administration of doxylamine and pyridoxine.

1. Introduction

Over the past few decades polymer microparticles have been considered as promising systems for nasal drug delivery. Due to their light weight they easily adhere on the nasal tissue and reach vast mucosal surface areas, which result in a high drug uptake (Deutel et al. 2015). Moreover, various mucoadhesive polymers have been used in order to prolong the contact of the formulation with the mucus membrane. In particular, chitosan is one of the most extensively investigated mucoadhesive polymers and has been widely employed in the preparation of drug systems for mucosal administration (Casettari and Illum 2014). Due to their biodegradability, low toxicity, biocompatibility and excellent mucoadhesive properties, chitosan nano- and microparticles have been considered as appropriate vehicles for drug delivery (Mohammed et al. 2017; Ali and Ahmed 2018).

Despite the advantages of chitosan microparticles as pharmaceutical formulations, one major challenge persists. Achieving sustained drug release especially when incorporating water-soluble drugs is of significant importance and could offer diminished drug toxicity and dosage frequency. Chitosan matrix swells quickly in water and often releases rapidly the incorporated drugs, unable to provide sustained drug delivery. A possible approach to make the polymer structure more rigid and suited for controlled drug release is to modify the chitosan structure by cross-linking of its chains. Depending on the nature of the cross-linker, the interactions forming the cross-linked structure are mainly ionic or covalent bonds. In the ionic (physical) cross-linking process, a network of ionic bridges between negatively charged components and the positively charged chitosan chains is formed. Covalent (chemical)

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cross-linking on the other hand can more effectively affect the physicochemical stability of the chitosan matrix since the cross-linking process is irreversible. The higher stability of such modified chitosan is based mainly on the covalent bonds, but other interactions (hydrogen or hydrophobic bonds) are also possible (Szymańska and Winnicka 2015). To date, one of the most common chemical cross-linkers of chitosan is glutaraldehyde. The primary amine ($-NH_2$) groups of chitosan provide a reaction site for covalent interaction with the aldehyde molecules, thus allowing polymer modification and achieving a stable cross-linked network (Desai and Park 2005; Gonçalves et al. 2005; Gupta and Jabrail 2006; Kulkarni et al. 2007; Ramachandran et al. 2011).

Although cross-linking of chitosan has been proposed by many authors as a possible approach to modify the drug release pattern, it could negatively influence the mucoadhesive characteristics of the polymer. The degree of cross-linking significantly influences polymer chain mobility, which is crucial for interpenetration and entanglement with the mucus gel. It is known that the more flexible the chains, the better interdiffusion and penetration of the polymer within the mucus network. As cross-linking density increases, polymer chain mobility decreases and hence the mucoadhesive strength is reduced (Andrews et al. 2009). Therefore, the amount of the cross-linker should be optimized and its influence on both release kinetics and mucoadhesion should be evaluated.

The aim of this study was to formulate mucoadhesive glutaraldehyde cross-linked chitosan microparticles as sustained release drug delivery systems *via* spray-drying technique and to investigate how the process of cross-linking affects their physical and chemical characteristics.

Another challenging aspect of this work was the simultaneous incorporation of two APIs in the microparticles. Doxylamine succinate (DOX) and pyridoxine hydrochloride (PYR) were used as model drugs. The combination has been approved on the pharmaceutical market for the treatment of nausea and vomiting with proven efficacy and safety during pregnancy (Slaughter et al. 2014). So far, this drug combination is available only in various oral dosage forms with modified drug release. However, the oral administration is associated with certain constraints and is not particularly suited when treating nausea and vomiting. The nasal cavity, on the other hand, is nowadays being dynamically explored for systemic delivery of various therapeutic agents including antiemetics such as metoclopramide (Gavini et al. 2008) and ondasetrone (Gungor et al. 2010).

The design of polymeric microcarriers with optimal bioadhesive and biopharmaceutical properties is a technological challenge; its successful realization would be a prerequisite for the formulation of a novel and convenient drug delivery system for nasal administration of doxylamine and pyridoxine.

2. Material and methods

2.1. Materials

Chitosan from shrimp shells (low-viscosity, degree of deacetylation > 70%), doxylamine succinate, pyridoxine hydrochloride, glutaraldehyde (50% aqueous solution), mucin (porcine stomach, type II) and Bradford reagent were purchased from Sigma Aldrich, USA. All solvents and reagents were of analytical grade.

2.2. Methods

2.2.1. Microspheres preparation

Microspheres were formulated by spray-drying technique. Solutions with concentration of chitosan 2% (w/v), doxylamine 1% (w/v) and pyridoxine 1% (w/v) were prepared in 2% (v/v) acetic acid. A chemical cross-linking reaction was carried out by adding glutaraldehyde (GA) dropwise into the resultant solutions under stirring (400 rpm) for 20 min (Gaspar et al. 2015). Three different concentrations of GA were used: 0.1, 0.5 and 1 mg/mL. The mixtures were spray-dried through a 0.7 mm nozzle using Büchi Mini Spray Dryer B-290 (Büchi Labortechnik AG, Flawil, Switzerland). The spray-drying parameters were previously optimized applying 3^2 factorial design and constant process settings were used: 140 °C inlet temperature, 160 mL/h pump rate, 600 L/h airflow rate and 95% aspiration (Katsarov et al. 2017a). Non-cross-linked microspheres were also prepared as a basis for comparison.

2.2.2. Scanning electron microscopy (SEM)

Scanning electron photographs of the formulated particle were taken using Philips SEM 515 (Philips, Eindhoven, The Netherlands). Small amounts of microspheres were fixed with double-sided adhesive on holders and were coated with a thin layer of gold in SC7620 Mini Sputter Coater (Quorum Technologies Ltd., UK). SEM was performed at 25 kV acceleration voltage and 5000× magnification.

2.2.3. Evaluation of production yield, size and particle size distribution

Particle production yield was calculated as the ratio of the weight of the obtained microspheres (W_1) to the weight of the spray-dried polymer and drugs (W_2) according to the following equation: *Yield* (%) = $(W_1 / W_2) \times 100$. Particles size distribution and median diameter were determined by laser diffraction technique using LS 13320 analyzer (Beckman Coulter, USA). Dry samples of 100 mg particles were analyzed with a Tornado system for powdered samples. All measurements were repeated three times.

2.2.4. UV spectrophotometric analysis

Simultaneous spectrophotometric quantitative determination of doxylamine and pyridoxine was carried out using Evolution 300 UV–visible spectrophotometer (Thermo Fisher Scientific, USA). The absorption spectra of the sample solutions were recorded in the wavelength range of 200–400 nm at 1.0 nm intervals, using VisionPRO™ software (Thermo Fisher Scientific). In order to separate the overlapping drug spectra, a multivariate chemometric method on the principle of partial least squares (PLS) was applied (Wold et al. 2001). A PLS2 model for the binary mixture DOX/PYR was constructed on the basis of 11 mixtures containing different concentrations of DOX and PYR (Katsarov et al. 2018). PLS2 calibration was carried out using the MVC1® MatLab toolbox (Olivieri et al. 2004).

2.2.5. Determination of drug content in the formulations

The amount of doxylamine and pyridoxine in the particles was extracted with phosphate buffered saline (PBS), pH 6.8 for 24 h at room temperature and was determined spectrophotometrically applying the developed PLS2 method. Drug loading (*DL*) (%) was calculated as the ratio of the determined amount of drug (mg) in the microspheres (W_d) to the weight (mg) of the microspheres (W_m) according to the following equation: DL (%) = $(W_d / W_m) \times 100$. Drug entrapment efficiency (*DEE*) (%) was calculated on the basis of the determined drug loading and the estimated theoretical drug loading (Saigal et al. 2013). All experiments were performed in triplicate, and the means of the obtained values were reported.

2.2.6. Fourier transformed infrared spectroscopy (FTIR)

Infrared spectra of the cross-linked microspheres were recorded on a Nicolet iS 10 FTIR spectrometer (Thermo Fisher Scientific, Pittsburgh, PA, USA), equipped with a diamond attenuated total reflection (ATR) accessory. The analysis was performed under the following conditions: 64 scans, resolution of 4 nm, and spectral range 4000–650 cm^{-1}.

2.2.7. X-ray powder diffraction (XRPD)

XRPD analysis of the cross-linked microspheres was performed in order to characterize the physical state of the loaded drugs. Diffractograms were recorded on a D2 Phaser X-ray powder diffractometer (Bruker AXS GmbH, Germany). Ni-filtered Cu radiation at

30 kV and 10 mA was used. The scan angle range (2θ) was 4–70°. Powder Diffraction database (ICDD) was used for phase analysis and interpretation of the results.

2.2.8. Microspheres swelling properties

Accurately weighed amount of microspheres was dispersed into 1 ml PBS (pH 6.8). At predetermined time intervals, the samples were centrifuged for 5 min in a mini centrifuge MPW-55 (Med. Instruments, Poland) at 5000 rpm. The supernatant was removed and the weight of the swollen microspheres was estimated. The swelling index (*SW*) was calculated as the percentage ratio of the swollen particle weight (W_t) to the initial weight of the particles (W_i) according to the following equation: $SW\ (\%) = (W_t / W_i) \times 100$ (El-Sherbiny et al. 2010). The experiments were performed in triplicate.

2.2.9. In vitro drug release

In vitro drug release studies were carried out in 400 ml PBS (pH 6.8) using a USP Apparatus I rotating basket (AT7 Sotax, Allschwil, Switzerland). Dialysis tubing (Sigma, MWCO 12000 Da) was used to prepare membrane bags with size 6 × 2.5 cm. Predetermined amounts of microparticles, equivalent to 30 mg drug content, were suspended into 1 mL PBS medium and transferred in the membrane bags. The bags were placed in the baskets of the dissolution apparatus. The basket rotation speed was kept constant at 50 rpm. The medium temperature was maintained at 37 ± 0.5 °C. Aliquots of 5 mL were withdrawn at determined time intervals, filtered (0.45 μm membrane filter) and assayed spectrophotometrically for drug content, applying the developed PLS2 method for analysis. The withdrawn aliquots were replaced with fresh medium. Each experiment was performed in triplicate (SD provided).

2.2.10. In vitro mucoadhesion study

Accurately weighed amount of microspheres was dispersed into 10 mL mucin aqueous solution (mucin Type II, extracted from porcine stomach) and stirred for 2 h at 50 rpm. The obtained suspension was centrifuged at 3000 rpm for 5 min using a Cencom centrifuge (J.P. Selecta, Spain). The amount of free mucin in the supernatant was assayed using Bradford colorimetric method (Han et al. 2012). The percentage of mucin adsorbed on the microparticles surface was calculated from the difference between the initial amount of mucin (M_0) and the determined free mucin (M_f) according to the following equation: *Adsorbed mucin* (%) = $[(M_0 - M_f) / M_0] \times 100$.

3. Results and discussion

3.1. Formulation of cross-linked microspheres

Glutaraldehyde, which is considered to be one of the most effective chitosan cross-linking agents, was chosen for the preparation of cross-linked microparticles. Its reaction with chitosan has been well-documented (Monteiro and Airoldi 1999). Technological approach based on a series of preliminary studies for chitosan cross-linking before spray-drying was applied. Three models were formulated by spray-drying of chitosan solutions to which different amounts of GA (concentrations 0.1 to 1 mg/mL) were added (Table 1). The colour of the resultant microspheres was accepted as an initial indicator for the degree of polymer cross-linking. The more intense the cross-linking, the darker and more saturated the particles colour was (Monteiro and Airoldi 1999). No visible changes in the colour of the microspheres was observed at GA concentration lower than 0.1 mg/mL. On the other hand, GA concentration higher than 1 mg/ml resulted in the formation of a chitosan system with semi-solid consistency, which was impossible to spray-dry.

3.2. Yield, drug loading, morphology and particle size

The formulated microparticles showed yields between 53% and 59%, which could be considered as substantially high. The spray-drying technique does not usually result in production yields above 60% due to powder sticking to the equipment's parts (nozzle, drying chamber, cyclone) (Fasiolo et al. 2018). After chitosan cross-linking, the spray-dried material adhered more slightly on the walls of the chamber. This resulted in more efficient separation and collection of the formed microparticles and hence in higher yields (56–59%) compared to the non-cross-linked model M.DOX.PYR (53% yield).

Table 1
Median particles diameter and production yield of doxylamin/pyridoxine loaded chitosan microparticles, cross-linked with different concentrations of glutaraldehyde (mean ± SD, $n = 3$).

Formulation	Glutaraldehyde concentration (mg/mL)	Particle median diameter (μm ± SD)	Production yield (% ± SD)
M.DOX.PYR	–	4.70 ± 0.32	53.63 ± 3.42
CR.DOX.PYR.1	0.1	4.10 ± 0.27	56.72 ± 4.84
CR.DOX.PYR.2	0.5	4.05 ± 0.37	58.80 ± 2.36
CR.DOX.PYR.3	1.0	4.03 ± 0.16	59.53 ± 2.38

The cross-linked particles showed high drug entrapment efficiency (80.47 to 94.25%) (Table 2). In all models DOX and PYR were incorporated in approximately equal amounts (*DL* for DOX: 20.12–23.01%; *DL* for PYR: 21.00–23.56%), which corresponded to the same therapeutic doses of the drugs (Slaughter et al. 2014). Increased cross-linker concentrations resulted in lower drug encapsulation efficiency. No remarkable differences in *DL* values of models CR.DOX.PYR.1 and CR.DOX.PYR.2, formulated with 0.1 and 0.5 mg/mL GA (46.57 and 46.24% respectively) were observed. However, model CR.DOX.PYR.3 showed the lowest *DL* (41.12%). GA concentration as high as 1 mg/ml limited the amount of incorporated DOX and PYR in the microspheres. Similar results have been observed by other authors and have been attributed to possible interaction between the cross-linker and the drug, resulting in a complex that could be spray dried separately from the microspheres (Gungor et al. 2010; Ventura et al. 2008). However, in our studies neither the X-ray powder diffraction, nor the scanning electron microscopy indicated the presence of such structures.

The obtained microparticles were with spherical shape and smooth surface. No visible pores, deformations or drug crystals were observed. SEM images of the models are presented in Fig. 1.

The particle size analysis showed that all cross-linked microspheres were of similar size (Table 1 and Fig. 2). Their median diameter (4.03–4.10 μm) was slightly smaller compared to the non-cross-linked model (4.70 μm). The cross-linking reaction generally affects the density of the polymer matrix. The use of GA is assumed to decrease the microsphere size, which is attributed to the formation of tighter

Table 2
Drug loading (*DL*) and drug entrapment efficiency (*DEE*) of non-cross-linked and cross-linked chitosan microspheres, loaded with doxylamine and pyridoxine (mean ± SD, $n = 3$).

Formulation	Drugs	*DL* (% ± SD)	*DEE* (% ± SD)
M.DOX.PYR	DOX + PYR (DOX; PYR)	47.83 ± 0.99 (23.40 ± 0.61; 24.43 ± 0.40)	95.67 ± 1.97 (93.60 ± 2.43; 97.73 ± 1.62)
CR.DOX.PYR.1	DOX + PYR (DOX; PYR)	46.57 ± 0.39 (23.01 ± 0.33; 23.56 ± 0.38)	93.14 ± 0.70 (92.03 ± 1.31; 94.25 ± 1.50)
CR.DOX.PYR.2	DOX + PYR (DOX; PYR)	46.24 ± 0.19 (22.96 ± 0.26; 23.28 ± 0.43)	92.48 ± 0.39 (91.84 ± 1.02; 93.12 ± 1.72)
CR.DOX.PYR.3	DOX + PYR (DOX; PYR)	41.12 ± 0.25 (20.12 ± 0.33; 21.00 ± 0.33)	82.24 ± 0.49 (80.47 ± 1.31; 84.01 ± 1.30)

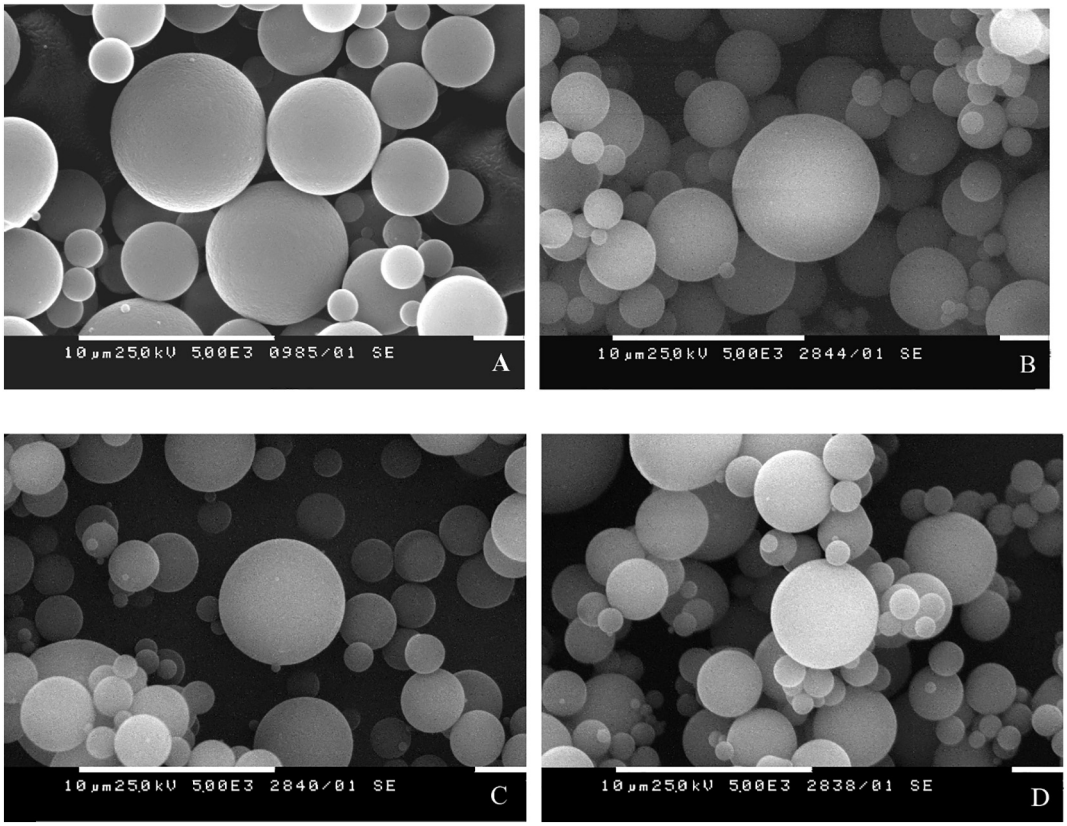

Fig. 1. SEM micrographs of non-cross-linked (A) M.DOX.PYR and cross-linked models (B) CR.DOX.PYR.1, (C) CR.DOX.PYR.2 and (D) CR.DOX.PYR.3 (5000×).

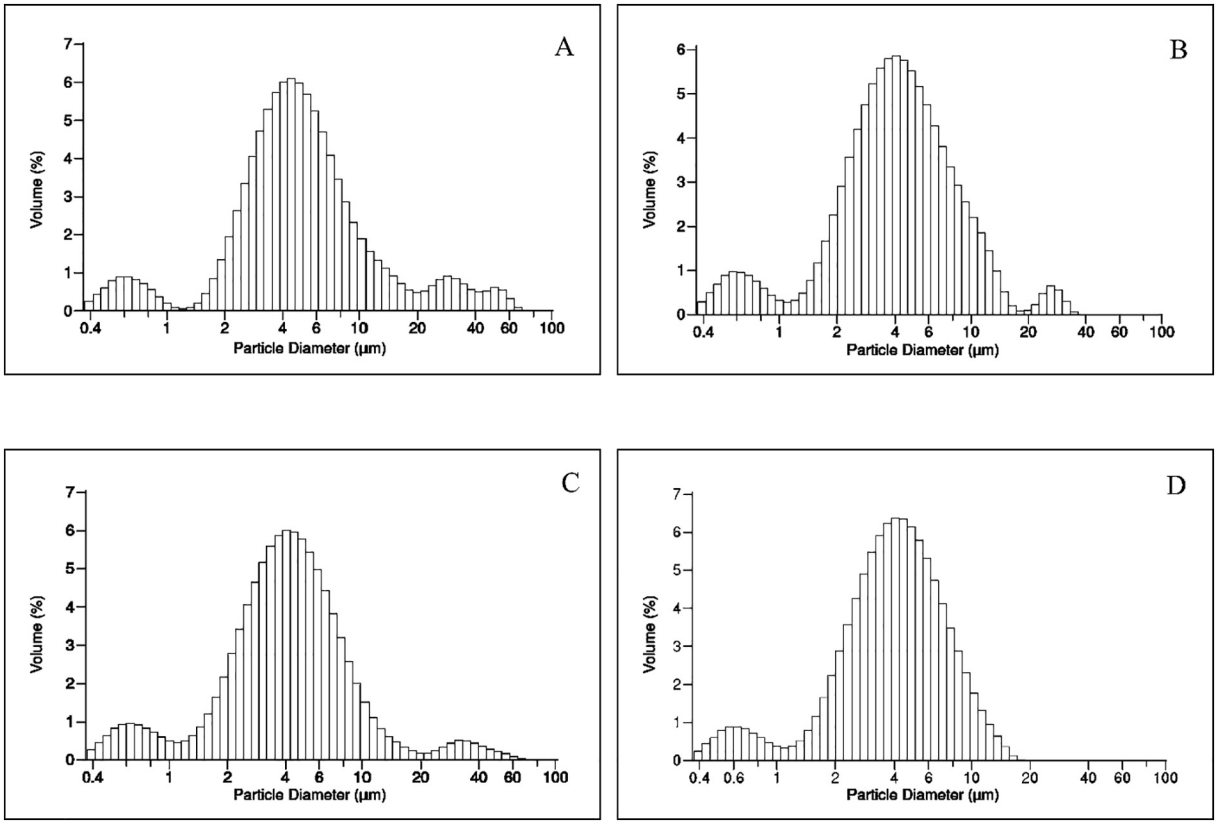

Fig. 2. Particle size distribution of non-cross-linked (A) M.DOX.PYR and cross-linked models (B) CR.DOX.PYR.1, (C) CR.DOX.PYR.2 and (D) CR.DOX.PYR.3.

chitosan structure. However, during cross-linking the viscosity of the polymer solution to be spray-dried increases, which is associated with spraying of larger droplets and further formation of larger particles (Patel and Patel 2014). The simultaneous influence of these two factors was probably the reason why the process of cross-linking did not considerably reduce the size of the resultant microspheres. That could be considered beneficial since the larger diameter determines a longer diffusion path of the incorporated drugs through the matrix and therefore a slower drug release. Moreover, the larger the microspheres, the more likely they are to be deposited in the nasal cavity and not inhaled in the lungs. The size of the obtained cross-linked microparticles (diameter from 0.4 to 60 μm) could be considered appropriate for nasal administration. According to the literature particles in the range 0.5–10 μm are deposited predominantly in the upper respiratory region of the nose, while smaller structures are more suited for pulmonary administration (Gungor et al. 2010).

3.3. FTIR and XRPD analysis

ATR-FTIR spectra of the obtained microparticles were recorded in order to evaluate the chemical interactions between chitosan and GA and any potential chemical modifications of the drugs as a result of the cross-linking process. The spectra of the cross-linked models are shown in Fig. 3.

In all of them were present characteristic peaks for the O—H groups of chitosan and PYR, N—H bonds, and C—H bonds of the aromatic rings (3000–3600 cm^{-1}). Peaks of stretching vibrations of C═O, C—O—C and CH_2—OH (1100–1025 cm^{-1}) and wagging C—H vibrations of the monosaccharide rings were also observed, which were typical for the chitosan spectrum (Katsarov et al. 2017b). The increased intensity of the band at 1633 cm^{-1}, detected only in the spectra of the cross-linked microspheres, could be considered as an indication for the cross-linking process. It was probably a result from the reaction of the amine groups of chitosan with GA and the formation of C═N bond (Gaspar et al. 2015).

The drug physical state is considered as an important parameter when evaluating the drug release from microparticle formulation. X-ray powder diffractograms of the cross-linked microspheres are presented in Fig. 4. They showed that all models had amorphous structure with a halo baseline (~4.10 Å), typical for chitosan. Small diffraction peaks were observed only in the diffractogram of model CR.DOX.PYR.3, but they were not characteristic for the incorporated drugs (Katsarov et al. 2017b).

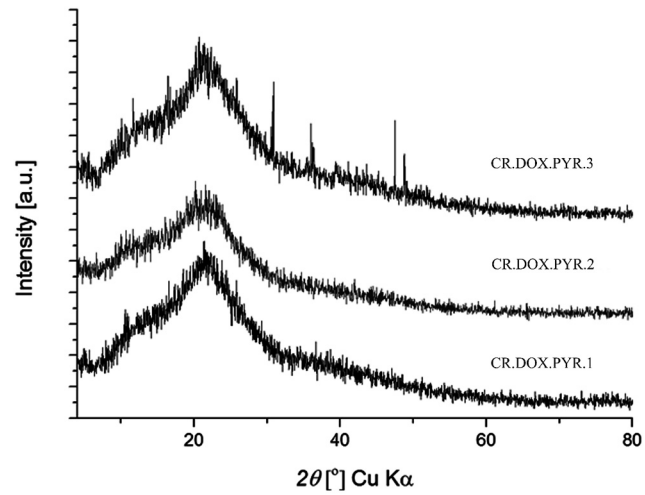

Fig. 4. X-ray powder diffractograms of glutaraldehyde cross-linked microparticles models CR.DOX.PYR.1, CR.DOX.PYR.2 and CR.DOX.PYR.3.

The absence of sharp peaks indicated that DOX and PYR had lost their crystalline structure, becoming amorphous during the spray drying process. The formation of amorphous structure is generally associated with faster rate of drug release (Tres et al. 2015). Since both DOX and PYR are very soluble in water the registered transformation should not affect considerably their dissolution profile.

3.4. Swelling properties

The polymer swelling capacity is an important factor, which influences the drug release from the chitosan matrix. The swelling profiles of non-cross-linked and cross-linked microparticles are presented in Fig. 5.

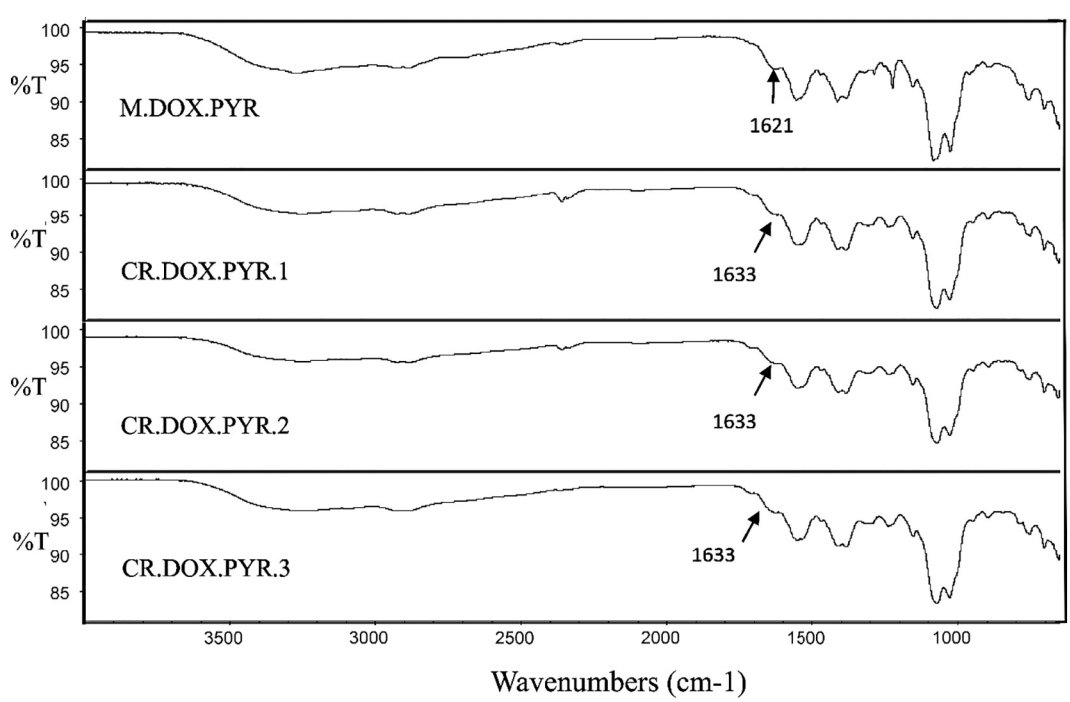

Fig. 3. FTIR-ATR spectra of non-cross-linked (M.DOX.PYR) and cross-linked microparticles models CR.DOX.PYR.1, CR.DOX.PYR.2 and CR.DOX.PYR.3.

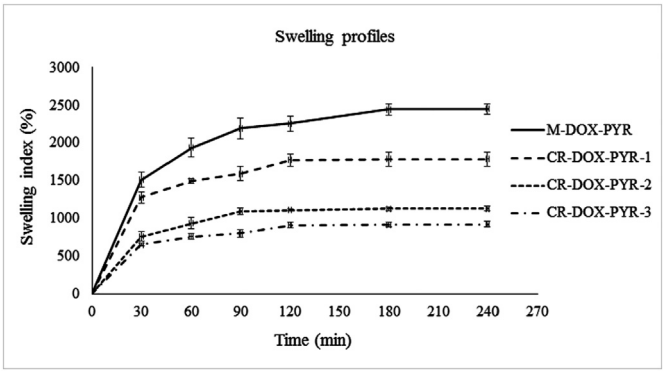

Fig. 5. Swelling profiles of non-cross-linked (M.DOX.PYR) and cross-linked microparticles models CR.DOX.PYR.1, CR.DOX.PYR.2 and CR.DOX.PYR.3 (n = 3).

They were based on the change in the swelling index of the microspheres over time within 4 h of contact with PBS pH 6.8.

A reduction in particle swelling was observed as a result of the cross-linking process. The results indicated that swelling capacity was closely associated with the level of cross-linking. Increasing the concentration of GA caused considerable restriction of particles swelling. Within the trial, model CR.DOX.PYR.1 reached a maximum swelling index of 1786%, while CR.DOX.PYR.2 and CR.DOX.PYR.3 models: 1132% and 924% respectively. The more tightly cross-linked the polymer matrix, the less its ability to absorb water. The strongly cross-linked chitosan chains are less flexible and have fewer free hydrophilic groups for hydrogen bonding with water molecules and therefore their swelling capacity is limited (Mirzaei et al. 2013). Polymer matrices with reduced swelling are expected to have slower drug release (Berger et al. 2004; Desai and Park 2005).

3.5. In vitro drug release

The drug release profiles of DOX and PYR from the cross-linked and non-cross-linked microsphere models are presented in Fig. 6.

The drug release from the cross-linked models, similarly to the non-cross-linked microspheres, was characterized with an initial burst effect. A biphasic release profile is typical for polymer drug delivery systems. Phase I is the rapid release of drug molecules located close to the surface or near the water layer and phase II is a slow release phase governed by slow drug diffusion through the polymer matrix (Kamaly et al. 2016). The observed initial burst release of the drugs from the microparticles could be associated with their high solubility in water and the extensive swelling of the polymer carrier. As already shown in Fig. 5, the microspheres swelled and increased their mass several times after the contact with an aqueous medium.

However, from the profiles shown in Fig. 6, it is clear that cross-linking of chitosan considerably reduced the drug burst release. With the increase in GA concentration from 0.1 to 1 mg/mL, a delay in the initial drug dissolution was observed. Generally, with greater cross-linker amounts a thicker polymer matrix is formulated, which swells to a lesser extent and releases the incorporated drugs more slowly (Raval et al. 2010).

The non-cross-linked microspheres (M.DOX.PYR) released 42.14/49.96% and 61.99/68.99% of DOX/PYR for the first 15 min and 30 min respectively, while the cross-linked models released 33.07/34.22% and 51.49/53.37% (CR.DOX.PYR.1), 12.50/13.18% and 33.05/34.01% (CR.DOX.PYR.2), 8.12/9.04% and 29.23/30.15% (CR.DOX.PYR.3) of the drugs for the same periods. In order to differentiate the burst effect from the second release phase, the release profiles of DOX and PYR after the first 15 min and after the first 30 min were fitted to different mathematical models. The calculated coefficients of determination (R^2) for each model are presented in Table 3.

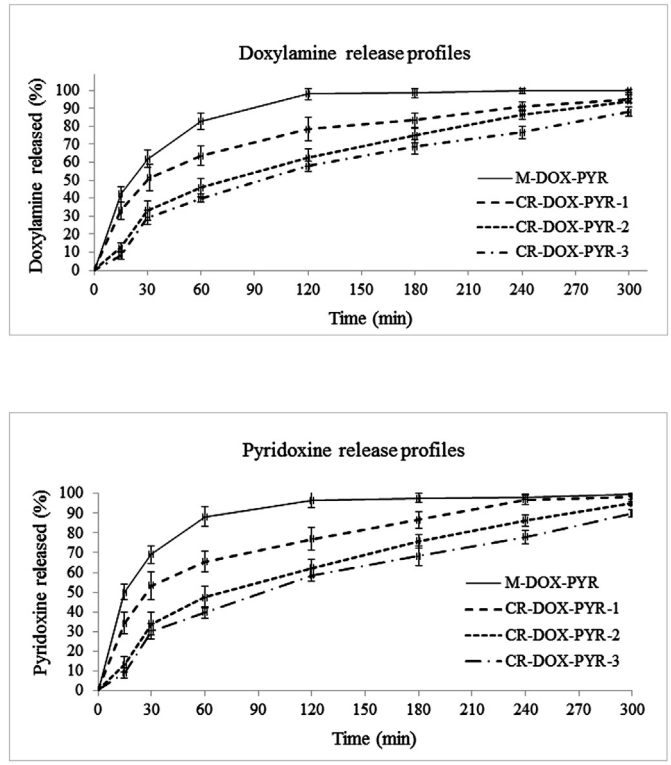

Fig. 6. Release profiles of doxylamine and pyridoxine from non-cross-linked (M.DOX.PYR) and cross-linked microparticles models CR.DOX.PYR.1, CR.DOX.PYR.2 and CR.DOX.PYR.3 (n = 3).

Table 3
Estimated coefficients of determination (R^2) for different kinetic models.

Kinetic model		CR.DOX.PYR.1		CR.DOX.PYR.2		CR.DOX.PYR.3	
		DOX	PYR	DOX	PYR	DOX	PYR
Zero order	*	0.8541	0.8760	0.9219	0.9216	0.9179	0.9265
	**	0.9203	0.9421	0.9721	0.9755	0.9741	0.9796
First order	*	0.9875	0.9664	0.9796	0.9670	0.9788	0.9691
	**	0.9893	0.9567	0.9734	0.9571	0.9752	0.9602
Higuchi	*	0.9450	0.9557	0.9813	0.9800	0.9774	0.9796
	**	0.9778	0.9852	0.9989	0.9991	0.9974	0.9961
Korsmeyer-Peppas	*	0.9587	0.9632	0.9357	0.9350	0.9027	0.9131
	**	0.9932	0.9944	0.9996	0.9991	0.9984	0.9964
Hixson-Crowell	*	0.9684	0.9771	0.9901	0.9875	0.9790	0.9797
	**	0.9888	0.9797	0.9968	0.9922	0.9914	0.9865

*, data from drug release in the period 15 min–300 min.
**, data from drug release in the period 30 min–300 min.

The fitted release profiles after 30 min showed higher R^2 values compared to these after 15 min. The drug release after the first 30 min gave high linearity (0.99) for all cross-linked models and therefore could be considered as the second release phase, after the initial burst effect. The second release phase was best described by the Korsmeyer-Peppas model, which showed the highest values for R^2 (from 0.9932 to 0.9996). This mathematical model has been often used to analyze drug release kinetics from polymer systems. It represents the linear relationship between the logarithm of the released drug (Log M_t / M_0) for a period of time and logarithm of that time (Log t) (Notario-Pérez et al. 2017).

The diffusion exponent (n), which is an indication of the drug transport mechanism through the polymer matrix, was also calculated for the second DOX and PYR release phase. It was determined according to the following equation: $M_t / M_0 = K{\cdot}t^n$, where M_t / M_0 was the released amount of drug for a period of time (t), and K is the kinetic

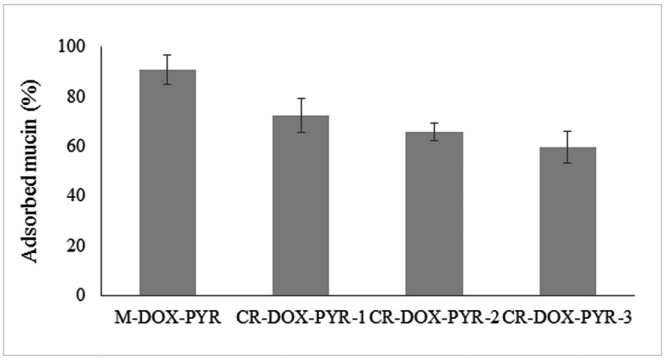

Fig. 7. *In vitro* adsorbed mucin (%) on non-cross-linked (M.DOX.PYR) and cross-linked microparticles models CR.DOX.PYR.1, CR.DOX.PYR.2 and CR.DOX.PYR.3 (mean ± SD, $n = 3$).

constant associated with the structural characteristics of the drug delivery system. All diffusion exponent values were below 0.5, which indicated that the drugs were released from the cross-linked microspheres by passive diffusion according to Fick's law.

By cross-linking, not only was the initial burst effect limited, but also a delayed drug release was subsequently achieved. In contrast to the non-cross-linked microspheres, which released almost the entire loading within 3 h, for the cross-linked models the release process lasted up to 5 h. In aqueous medium the denser polymer matrix of the cross-linked microparticles is likely to form a gel layer with higher viscosity, which slows up the diffusion of the drugs from the gel core and leads to prolonged drug release (Berger et al. 2004).

3.6. Mucoadhesion

The results from the *in vitro* mucoadhesion study (Fig. 7) showed that the largest amount of mucin (91%) was adsorbed on the non-cross-linked microspheres. That was expected due to the presence of free positively charged amino groups in the chitosan structure, which generally interact with the negatively charged mucin (Han et al. 2012). As a result of the chemical cross-linking process some of these groups are engaged in bonds with glutaraldehyde and therefore the mucoadhesive properties of the cross-linked microparticles are reduced (Genta et al. 1998). Our results confirmed that increased GA concentration led to weaker mucoadhesion. However, despite being cross-linked, the models CR.DOX.PYR.1, CR.DOX.PYR.2 and CR.DOX.PYR.3 also showed relatively high percentages of adsorbed mucin (72, 65 and 60% respectively) which led to the assumption that both cross-linked and non-cross-linked microspheres had satisfactory mucoadhesive properties.

4. Conclusion

Chemically cross-linked chitosan microparticles were formulated as potential carriers for sustained release of doxylamine and pyridoxine. The cross-linking process with glutaraldehyde in concentrations 0.1–1.0 mg/mL successfully limited the microparticle swelling and prolonged the *in vitro* drug release for 5 h. The cross-linked formulations demonstrated high drug entrapment efficiency and good mucoadhesive properties. The obtained mucoadhesive microsphere capable of sustained drug release are a perspective ground for further development of novel delivery systems for nasal administration of doxylamine and pyridoxine.

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